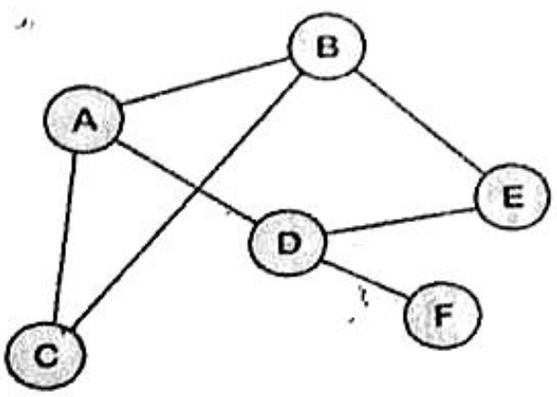
B.Tech (Computer Engineering)
(V Semester) 1st Sessional Exam: 2026-27
Data Analytics (Hons)
CSH– 511

Time: 1 hr

Max Marks : 12

Note: Attempt all questions

Q1 CO1	A. What are the 4 major tasks involved in data preprocessing and briefly define each. B. Explain the concept of "Garbage In, Garbage Out" C. List the key elements of data quality	[2+1+1] BL 1, 2
Q2 CO2	Compute the following: **(a)** Degree Centrality of all 6 nodes **(b)** Closeness Centrality of all 6 nodes **(c)** Betweenness Centrality of nodes A and D. 	[1+1+2] BL 3
Q3 CO2	A security system wants to compress 2 signals into a single Risk Score using Linear Discriminant Analysis (LDA). • Feature 1 (x1): Failed Login Attempts • Feature 2 (x2): Login Speed Anomaly (scale 1–10) • Classes: 1 = Genuine User, 2 = Bot Attack	[2+1+1] BL 3, 5

Attempt ID (Object)	Failed Logins	Speed Anomaly	Status
1	1	2	1 (Genuine)
2	2	1	1 (Genuine)
3	7	8	2 (Bot)
4	8	6	2 (Bot)

a) Determine the LDA axis's weight vector (w)
b) Compute the Risk Score for each of the 4 attempts listed above.
c) A new login attempt is recorded with:
x1=4 (Failed Logins), x2=9 (Speed Anomaly)
Based on the threshold (decision threshold = 665), classify this attempt as Genuine User or Bot Attack.